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EXTENDED ABSTRACT

A Machine Learning approach to study the ocean-atmosphere interactions in CESM2 for the North Atlantic

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1. Introduction and objectives

Over the North Atlantic, the first mode of atmospheric variability during winter is a dipolar surface pressure anomaly oscillating between a positive and a negative phase referenced as the North Atlantic Oscillation (NAO)[1]. The NAO can be influenced by local sea surface temperature (SST) anomalies through direct heat exchanges and non-linear feedback due to atmospheric dynamics[2]. In addition, remote SST anomalies located in the tropical regions of the Indian and Pacific Oceans can impact the NAO via various teleconnections that differ in early and late winter[3–5]. Although these relationships are difficult to assess as they are weak and non linear, they represent an interesting source of atmospheric predictability due to the longer persistence of SST anomalies compared to the chaotic nature of the atmosphere.

Learning the non-linear relationships between SST and NAO is challenging, as observational data have a limited length. Climate model simulations can thus provide much larger datasets, allowing the use of machine learning algorithms for their understanding. In this context, this research aims to investigate the relationships between sea level pressure (SLP) and SST anomalies by predicting the NAO using only past SST fields, handling the outputs of an unprecedentedly large dataset from the Community Earth System Model (CESM2) climate model simulations.

2. Methods and data

SST and SLP data are extracted from 89 out of the 100 ensemble members of the CESM2 historical simulations, which represent the climate and its evolution due to external forcings over the 1850-2014 period[6]. SLP fields are cropped over the North Atlantic region (20°N-70°N, 100°W-40°E) and SST fields are cropped between latitudes 15°S and 70°N. For both fields, the climatology is removed by subtracting daily mean values computed for the entire period studied. A principal component analysis (PCA) is performed on the SLP fields to extract their first principal component (PC1), explaining 24% of the total variance. The spatial pattern associated with the PC1 captures the NAO geographical structure, while the standardized PC1 is calculated as an NAO index.

SLP predictions are obtained using a Convolutional Neural Network (CNN) model taking as input three SST fields with lags of 35, 65 and 95 days and as output the NAO index. The CNN is composed of three convolutional blocks and a last fully connected layer. As tropical teleconnections impact the

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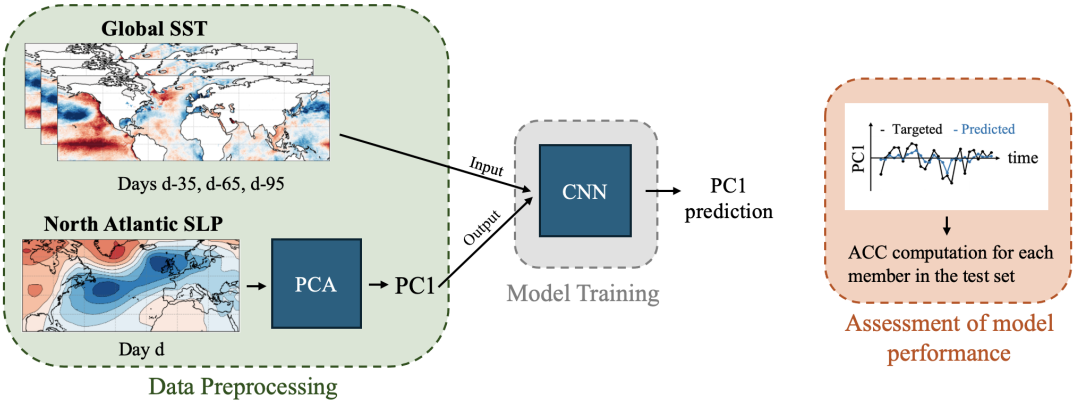

Figure 1. *Global schematic of the research.*

NAO differently in early and late winter[5], two separate models are trained in November/December and January/February. Both models are trained with 9 ensemble members and validated on a single member, while 79 additional members are retained as an independent test set.

3. Preliminary results

Before analyzing the results, the daily predictions are monthly averaged. An anomaly correlation coefficient (ACC) is then computed for each member of the test set between the predicted and targeted NAO index time series, separately for November, December, January and February. The statistical significance of the ACC is assessed using the bootstrap method with a 95% significance level. In February(January), preliminary results indicate that 97%(70%) of the test members have a statistically significant ACC. The average value of these statistically significant ACC is 0.25 in January and 0.30 in February. In November and December less than 5% of the test members present a significant ACC, highlighting that the model fails to generalize to the test set in early winter.

As signs of overfitting were present during training, expanding the training set could help mitigate this issue. Furthermore, applying XAI techniques could offer interesting future perspectives for a better understanding of the sources of predictability.

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